Morphology of Flowering Plants

Exam Standard: NEET UG | **Scope:** Root, Stem, Leaf, Inflorescence, Flower, Fruit, Seed, Floral Formula & Diagram, and Family Solanaceae.

1. Learning Objectives & Unit Overview

Before experimental biology and physiology were established as distinct branches of biology, naturalists studied living organisms primarily through observation—first with the naked eye and later using magnifying lenses and microscopes. These detailed descriptions of gross external (morphological) and internal (anatomical) structural features, along with observable living phenomena, formed the foundation of natural history. Later, reductionist biology utilized these structural descriptions to frame meaningful research questions in plant physiology and evolutionary biology.

Key Learning Objectives for NEET UG:

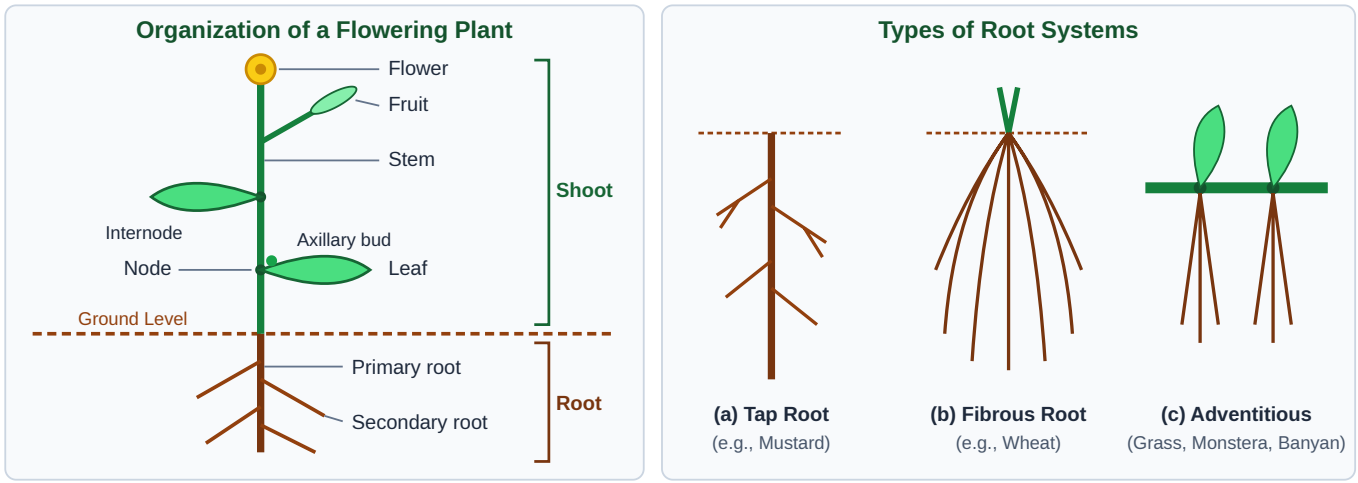
- Master standard technical terms and definitions required to classify and understand angiosperms (flowering plants).
- Differentiate between tap, fibrous, and adventitious root systems, and identify the structural regions of a growing root tip.
- Understand the morphological features of the stem and leaf, including leaf parts, venation, simple vs. compound leaves, and phyllotaxy.
- Compare racemose and cymose inflorescences based on shoot apex behaviour and floral arrangement.
- Analyze flower structure, symmetry, insertion of floral whorls on the thalamus, aestivation, cohesion/adhesion of stamens, and placentation.
- Distinguish between the morphological features of drupe fruits, dicotyledonous seeds, and monocotyledonous seeds.
- Construct and interpret floral formulas and floral diagrams (specifically Brassicaceae and Solanaceae).

Pioneer in Plant Structure: Katherine Esau (1898–1997)

- **Early Life & Education:** Born in Ukraine in 1898; studied agriculture in Russia and Germany and received her doctorate in 1931 in the United States.
- **Key Scientific Finding:** Reported in her early publications that the **curly top virus** spreads through a plant via the food-conducting or **phloem tissue**.
- **Landmark Publications:**
  - *Plant Anatomy* (published in 1954): Took a dynamic, developmental approach to plant structure and brought about a worldwide revival of the discipline.
  - *The Anatomy of Seed Plants* (published in 1960): Referred to as the **"Webster's of plant biology"** because of its encyclopaedic nature.
- **Honours:** Elected to the National Academy of Sciences in 1957 (becoming the sixth woman to receive that honour) and awarded the National Medal of Science by President George Bush in 1989.
- **Legacy:** When she died in 1997, Peter Raven (Director of Anatomy and Morphology, Missouri Botanical Garden) remembered that she "absolutely dominated" the field of plant biology even at the age of 99.

2. The Angiosperm Plant Body & The Root

Despite showing immense diversity in external structure (morphology), all angiosperms are characterised by the presence of **roots, stems, leaves, flowers, and fruits**. The underground part of the flowering plant forms the **root system**, while the portion above the ground forms the **shoot system**.



Tutorial Diagram 1: Parts of a typical flowering plant and comparison of Tap, Fibrous, and Adventitious root systems.

2.1 Types of Root Systems

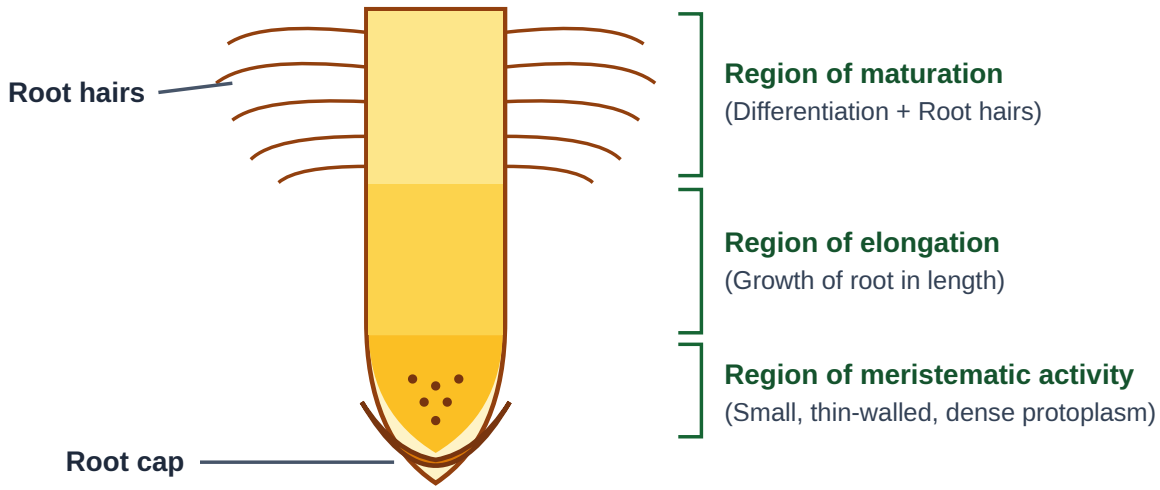
Root System	Origin & Developmental Features	Standard Examples
Tap Root System	Found in the majority of **dicotyledonous plants**. Direct elongation of the **radicle** forms the **primary root**, which grows inside the soil and bears lateral roots of several orders (secondary, tertiary, etc. roots).	Mustard plant (*Brassica*)
Fibrous Root System	Found in **monocotyledonous plants**. The primary root is short-lived and is replaced by a large number of roots originating from the **base of the stem**.	Wheat plant (*Triticum*)
Adventitious Roots	Roots that arise from parts of the plant **other than the radicle**.	Grass, *Monstera*, and the banyan tree

Main Functions of the Root System:

1. Absorption of water and minerals from the soil.
2. Providing a proper anchorage to the plant parts.
3. Storing reserve food material (in modified roots).
4. Synthesis of **plant growth regulators (PGRs)**.

2.2 Regions of the Root-Tip

Moving from the apex of the root upwards (proximally towards the stem base), four distinct structural zones are observed:



Tutorial Diagram 2: Structural regions of the root-tip from apex to maturation zone.

- **Root Cap:** A thimble-like structure covering the root apex. It protects the tender apex of the root as it makes its way through the soil.
- **Region of Meristematic Activity:** Located a few millimetres above the root cap. Cells of this region are **very small, thin-walled, and have dense protoplasm**. They divide repeatedly.
- **Region of Elongation:** Located *proximal* to the meristematic region. Cells undergo rapid elongation and enlargement and are **responsible for the growth of the root in length**.
- **Region of Maturation:** Located *proximal* to the region of elongation. Cells of the elongation zone gradually differentiate and mature here. Some of the **epidermal cells** of this region form very fine, delicate, thread-like structures called **root hairs**, which absorb water and minerals from the soil.

3. The Stem and The Leaf

3.1 The Stem

The stem is the ascending part of the plant axis bearing branches, leaves, flowers, and fruits. It develops from the **plumule** of the embryo of a germinating seed.

- **Distinguishing Features of a Stem (vs. Root):** Presence of **nodes** (regions where leaves are borne) and **internodes** (portions between two nodes), presence of terminal or axillary buds, multicellular hair, and positively phototropic nature.
- **Appearance:** Generally green when young and later often becomes woody and dark brown.
- **Functions:** Spreading out branches bearing leaves, flowers, and fruits; conducting water, minerals, and photosynthates. Some stems also perform functions of food storage, support, protection, and vegetative propagation.

3.2 The Leaf

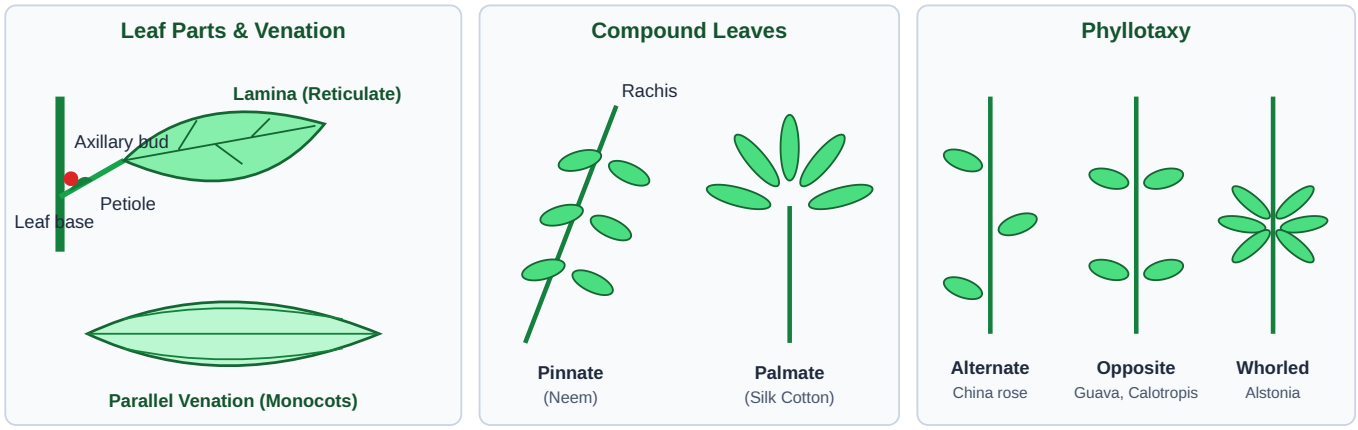
The leaf is a lateral, generally flattened structure borne on the stem. It develops **exogenously at the node** and bears a bud in its axil (the **axillary bud**, which later develops into a branch). Leaves originate from **shoot apical meristems** and are arranged in an **acropetal order** (younger leaves towards the apex, older towards the base). They are the most important vegetative organs for photosynthesis.

Three Main Parts of a Typical Leaf:

1. **Leaf Base (Hypopodium):** Attaches the leaf to the stem.
   - **Stipules:** Two lateral small leaf-like structures that may be borne on the leaf base.
   - **Sheathing Leaf Base:** In **monocotyledons**, the leaf base expands into a sheath covering the stem partially or wholly.

- **Pulvinus:** In some **leguminous plants**, the leaf base may become swollen and is called the *pulvinus*.

2. **Petiole (Mesopodium):** Helps hold the leaf blade to light. Long, thin, flexible petioles allow leaf blades to flutter in the wind, thereby cooling the leaf and bringing fresh air to the leaf surface.
3. **Lamina or Leaf Blade (Epipodium):** The green expanded part of the leaf with veins and veinlets. Usually has a prominent middle vein called the **midrib**. Veins provide rigidity to the leaf blade and act as channels of transport for water, minerals, and food materials.



Tutorial Diagram 3: Leaf structure, venation types, compound leaves, and phyllotaxy patterns.

3.3 Venation, Types of Leaves, and Phyllotaxy

Feature	Types & Definitions	Key Examples
Venation (Arrangement of veins and veinlets in the lamina)	• **Reticulate:** Veinlets form a network. • **Parallel:** Veins run parallel to each other within a lamina.	• Reticulate: Generally **dicotyledons**. • Parallel: Most **monocotyledons**.
Simple vs. Compound Leaves (Note: An axillary bud is present in the axil of the petiole in *both* simple and compound leaves, but **never** in the axil of leaflets)	• **Simple Leaf:** Lamina is entire, or when incised, the incisions do not touch the midrib. • **Pinnately Compound:** Incisions reach the midrib, breaking it into leaflets borne on a common axis called the **rachis** (which represents the midrib). • **Palmately Compound:** Leaflets are attached at a common point, i.e., at the **tip of the petiole**.	• Pinnately compound: **Neem** • Palmately compound: **Silk cotton**
Phyllotaxy (Pattern of arrangement of leaves on the stem or branch)	• **Alternate:** A single leaf arises at each node in an alternate manner. • **Opposite:** A pair of leaves arise at each node and lie opposite to each other. • **Whorled:** More than two leaves arise at a node and form a whorl.	• Alternate: **China rose, mustard, sunflower** • Opposite: ***Calotropis*, guava** • Whorled: ***Alstonia***

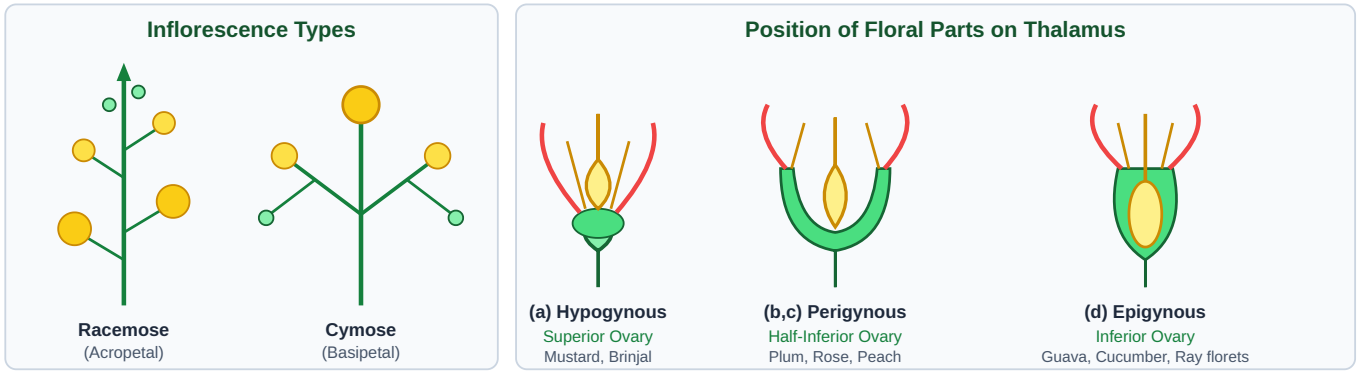
4. The Inflorescence and The Flower

4.1 The Inflorescence

A **flower is a modified shoot** wherein the shoot apical meristem changes to a **floral meristem**. Internodes do not elongate and the axis gets condensed. The apex produces different kinds of floral appendages laterally at successive nodes instead of leaves. When a shoot tip transforms into a flower, it is always **solitary**.

The arrangement of flowers on the floral axis is termed **inflorescence**. Depending on whether the apex gets developed into a flower or continues to grow, two major types are defined:

- **Racemose Inflorescence:** The main axis **continues to grow** (indeterminate/unlimited growth), and flowers are borne laterally in an **acropetal succession** (older flowers at the base, younger floral buds towards the growing apex).
- **Cymose Inflorescence:** The main axis **terminates in a flower**, hence is **limited in growth** (determinate). Flowers are borne in a **basipetal order** (oldest flower at the apex/centre, younger buds towards the base/periphery).



Tutorial Diagram 4: Racemose vs. Cymose inflorescence and Hypogynous, Perigynous, and Epigynous flowers.

4.2 General Terminology of the Flower

The flower is the reproductive unit in angiosperms, meant for sexual reproduction. A typical flower has four whorls arranged successively on the swollen end of the stalk or **pedicel**, called the **thalamus** or **receptacle**:

- **Accessory Organs:** Calyx and Corolla. (In some flowers like **lily**, the calyx and corolla are not distinct and are termed **perianth**; individual members are tepals).
- **Reproductive Organs:** Androecium (male) and Gynoecium (female).
- **Sexuality:** A flower having both androecium and gynoecium is **bisexual**; a flower having either only stamens or only carpels is **unisexual**.
- **Merosity:** A flower may be **trimerous, tetramerous, or pentamerous** when the floral appendages are in multiples of 3, 4, or 5, respectively.
- **Bracts:** Flowers with bracts (reduced leaf found at the base of the pedicel) are called **bracteate** (Br), and those without bracts are **ebracteate** (Ebr).

Floral Symmetry:

Symmetry Type	Symbol	Definition	Examples
Actinomorphic (Radial symmetry)	$\oplus$	Can be divided into two equal radial halves in **any radial plane** passing through the centre.	Mustard, *Datura*, chilli
Zygomorphic (Bilateral symmetry)	$\%$	Can be divided into two similar halves **only in one particular vertical plane**.	Pea, gulmohur, bean, *Cassia*
Asymmetric (Irregular)	—	Cannot be divided into two similar halves by any vertical plane passing through the centre.	*Canna*

Position of Floral Whorls on Thalamus:

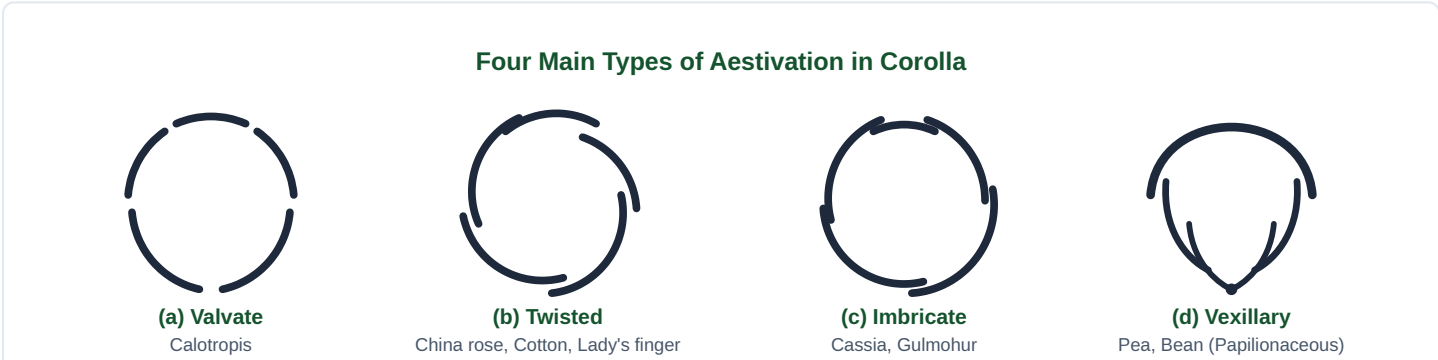
Flower Type	Ovary Position	Symbol	Structural Description	Examples
Hypogynous	**Superior**	$\underline{\mathrm{G}}$	Gynoecium occupies the highest position while other parts are situated below it.	Mustard, china rose, brinjal
Perigynous	**Half-inferior**	$\mathrm{G}-$	Gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level.	Plum, rose, peach
Epigynous	**Inferior**	$\overline{\mathrm{G}}$	Margin of thalamus grows upward enclosing the ovary completely and getting fused with it; other parts arise above the ovary.	Guava, cucumber, and ray florets of sunflower

5. Detailed Structure of Floral Whorls

5.1 Calyx & Corolla, and Aestivation

- **Calyx (Symbol K):** Outermost whorl; members are called **sepals**. Generally green, leaf-like, and protect the flower in the bud stage. May be **gamosepalous** (sepals united) or **polysepalous** (sepals free).
- **Corolla (Symbol C):** Composed of **petals**, usually brightly coloured to attract insects for pollination. May be **gamopetalous** (petals united) or **polypetalous** (petals free). Shape may be tubular, bell-shaped, funnel-shaped, or wheel-shaped.

Aestivation: The mode of arrangement of sepals or petals in a floral bud with respect to the other members of the same whorl is known as *aestivation*.



Tutorial Diagram 5: Aestivation patterns in floral buds—(a) Valvate, (b) Twisted, (c) Imbricate, and (d) Vexillary.

- **Valvate:** Sepals or petals in a whorl just touch one another at the margin, without overlapping. Example: ***Calotropis*** (also seen in mustard calyx/corolla and Solanaceae calyx/corolla).
- **Twisted:** One margin of the appendage overlaps that of the next one and so on in a regular direction. Examples: **China rose, lady's finger (okra), and cotton**.
- **Imbricate:** Margins of sepals or petals overlap one another, but **not in any particular direction**. Examples: ***Cassia* and gulmohur**.
- **Vexillary (or Papilionaceous):** Out of five petals, the largest posterior petal (**standard**) overlaps the two lateral petals (**wings**), which in turn overlap the two smallest anterior petals (**keel**). Written in floral formula as $\mathrm{C}_{1+2+(2)}$. Examples: **Pea and bean**.

5.2 Androecium (Symbol A)

Composed of **stamens** (male reproductive organs). Each stamen consists of a stalk or **filament** and an **anther**. Each anther is usually **bilobed**, and each lobe has two chambers, the **pollen-sacs**, where pollen grains are produced. A sterile stamen is called a **staminode**.

- **Adhesion of Stamens (Union with other floral whorls):**
  - **Epipetalous:** Stamens attached to the petals, e.g., **brinjal** $(\overline{\mathrm{C\ A}})$.

- **Epiphyllous:** Stamens attached to the perianth, e.g., flowers of **lily** ($\overline{\mathrm{P\ A}}$).

- **Cohesion of Stamens (Union among stamens themselves):**
  - **Polyandrous:** Stamens remain free.
  - **Monoadelphous:** Stamens united into **one bunch or one bundle**, e.g., **china rose**.
  - **Diadelphous:** Stamens united into **two bundles**, e.g., **pea**.
  - **Polyadelphous:** Stamens united into **more than two bundles**, e.g., ***Citrus***.

- **Variation in Filament Length:** There may be a variation in the length of filaments within a single flower, as in ***Salvia*** **and mustard**.

5.3 Gynoecium (Symbol G) and Placentation

Gynoecium is the female reproductive part of the flower and is made up of one or more **carpels**. A carpel consists of three parts:

1. **Ovary:** Enlarged basal part containing one or more ovules attached to a flattened, cushion-like **placenta**.
2. **Style:** Elongated tube lying on the ovary that connects the ovary to the stigma.
3. **Stigma:** Usually at the tip of the style; acts as the receptive surface for pollen grains.

When more than one carpel is present, they may be free—called **apocarpous** (e.g., **lotus and rose**)—or fused together—called **syncarpous** (e.g., **mustard and tomato**). After fertilisation, ovules develop into seeds and the ovary matures into a fruit.

Placentation: The arrangement of ovules within the ovary is known as *placentation*.



Tutorial Diagram 6: Five major types of placentation—(a) Marginal, (b) Axile, (c) Parietal, (d) Free central, and (e) Basal.

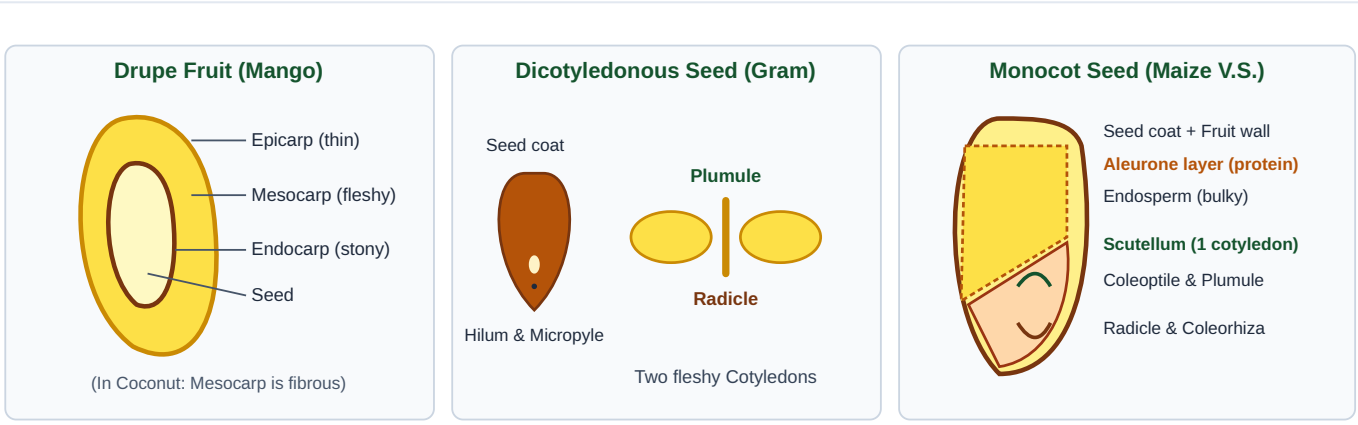
Placentation Type	Structural Description	Standard Examples
Marginal	Placenta forms a ridge along the **ventral suture** of the ovary, and ovules are borne on this ridge forming two rows.	Pea
Axile	Placenta is axial and the ovules are attached to it in a **multilocular ovary**.	China rose, tomato, lemon
Parietal	Ovules develop on the inner wall of the ovary or on the peripheral part. Ovary is one-chambered initially, but becomes **two-chambered due to the formation of a false septum (replum)**.	Mustard, *Argemone*
Free Central	Ovules are borne on the central axis and **septa are absent** (unilocular ovary).	*Dianthus*, Primrose
Basal	Placenta develops at the **base of the ovary** and a **single ovule** is attached to it.	Sunflower, marigold

6. The Fruit and The Seed

6.1 The Fruit

The fruit is a characteristic feature of flowering plants. It is a mature or ripened ovary, developed after fertilisation. If a fruit is formed **without fertilisation of the ovary**, it is called a **parthenocarpic fruit**.

- **Pericarp (Fruit Wall):** Generally, a fruit consists of a wall or **pericarp** and seeds. The pericarp may be dry or fleshy. When thick and fleshy, it is differentiated into:
  - Outer **epicarp**
  - Middle **mesocarp**
  - Inner **endocarp**
- **Drupe (Mango and Coconut):**
  - Develops from **monocarpellary superior ovaries** and is **one-seeded**.
  - In **Mango**: Pericarp is well-differentiated into an outer thin **epicarp** (skin), a middle fleshy **edible mesocarp**, and an inner **stony hard endocarp**.
  - In **Coconut** (also a drupe): The **mesocarp is fibrous** (coir), and the endocarp is stony hard enclosing the seed.



Tutorial Diagram 7: Structure of a Drupe (Mango), a Dicotyledonous Seed (Gram), and a Monocotyledonous Seed (Maize).

6.2 Comparison of Dicotyledonous and Monocotyledonous Seeds

After fertilisation, ovules develop into seeds. A seed consists of a **seed coat** and an **embryo** (made up of a radicle, an embryonal axis, and one or two cotyledons).

Feature	Dicotyledonous Seed	Monocotyledonous Seed (e.g., Maize)
Seed Coat	Has two distinct layers: outer **testa** and inner **tegmen**. Bears a scar called the **hilum** (attachment point to fruit) and a small pore above the hilum called the **micropyle**.	Membranous and generally **fused with the fruit wall** (pericarp).
Cotyledons	**Two cotyledons**, often fleshy and full of reserve food materials (as in gram and pea).	**One large, shield-shaped cotyledon** known as the **scutellum** (as in wheat, maize).
Embryonal Axis & Sheaths	Radicle and plumule are present at the two ends of the embryonal axis without protective sheaths.	Short axis with a **plumule** enclosed in a sheath called **coleoptile**, and a **radicle** enclosed in a sheath called **coleorhiza**.
Endosperm	Usually **non-endospermous** (endosperm absent in mature seeds, e.g., **bean, gram, pea**). *Exception:* **Castor** is an **endospermic** dicot seed.	Generally **endospermic** (bulky endosperm stores food, separated from the embryo by a proteinous **aleurone layer**). *Exception:* **Orchids** have **non-endospermic** monocot seeds.

7. Semi-Technical Description & Plant Families

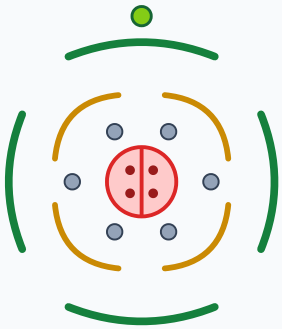
7.1 Symbols Used in Floral Formula

A flowering plant is described in a definite sequence: habit, vegetative characters (root, stem, leaves), floral characters (inflorescence and flower parts), followed by a **floral diagram** and a **floral formula**.

Symbol	Meaning	Symbol	Meaning
Br	Bracteate (with bract)	K	Calyx (sepals)
$\oplus$	Actinomorphic (radial symmetry)	C	Corolla (petals)
$\%$	Zygomorphic (bilateral symmetry)	P	Perianth (tepals)
$\mars$ (♂)	Male (staminate) flower	A	Androecium (stamens)
$\female$ (♀)	Female (pistillate) flower	G	Gynoecium (carpels)
$\hermaphrodite$ (⚥)	Bisexual flower	$\underline{\mathrm{G}}$ / $\overline{\mathrm{G}}$	Superior ovary ($\underline{\mathrm{G}}$) / Inferior ovary ($\overline{\mathrm{G}}$)
( )	Cohesion (fusion within a whorl)	$\overbrace{\mathrm{C\ A}}$	Adhesion (line drawn above symbols of connected whorls)

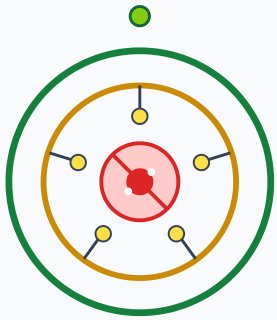
Key Features of a Floral Diagram: Provides information about the number of parts of a flower, their arrangement (aestivation), and the relation they have with one another. The position of the **mother axis** with respect to the flower is represented by a **dot on the top** of the floral diagram. Calyx, corolla, androecium, and gynoecium are drawn in successive concentric whorls, with calyx outermost and gynoecium in the centre.

Mustard (Family: Brassicaceae)



Formula: ⊕ ⚥ K 2+2 C 4 A 2+4 G(2) [Superior]

Solanum nigrum (Family: Solanaceae)



Formula: ⊕ ⚥ K(5) C(5) A5 [Epipetalous] G(2) [Superior]

Tutorial Diagram 8: Floral diagrams and formulas of Mustard (Brassicaceae) and Solanum nigrum (Solanaceae).

Mustard (Family: Brassicaceae) Floral Formula:

$$\oplus\ \hermaphrodite\ \mathrm{K}_{2+2}\ \mathrm{C}_{4}\ \mathrm{A}_{2+4}\ \underline{\mathrm{G}}_{(2)}$$

Interpretation: Actinomorphic, bisexual, 4 free sepals in two whorls of 2 each (K_{2+2}), 4 free petals (C_{4}), 6 free stamens arranged in two whorls of 2 outer short and 4 inner long (A_{2+4}), bicarpellary syncarpous superior ovary ($\underline{\mathrm{G}}_{(2)}$).

7.2 Family Solanaceae (The 'Potato Family')

A large family widely distributed in tropics, subtropics, and even temperate zones. Standard representative plant: *Solanum nigrum* (makoi).

- **Habit:** Mostly herbs, shrubs, and rarely small trees.
- **Stem:** Herbaceous, rarely woody, aerial; erect, cylindrical, branched, solid or hollow, hairy or glabrous; **underground stem in potato (*Solanum tuberosum*)**.
- **Leaves:** Alternate, simple, rarely pinnately compound, **exstipulate**; venation **reticulate**.
- **Inflorescence:** Solitary, axillary, or **cymose as in *Solanum***.
- **Flower:** Bisexual ($\hermaphrodite$), actinomorphic ($\oplus$).
- **Calyx:** Sepals five, united ($\mathrm{K}_{(5)}$), **persistent** (remains attached to the fruit, as in brinjal/tomato), valvate aestivation.
- **Corolla:** Petals five, united ($\mathrm{C}_{(5)}$), valvate aestivation.
- **Androecium:** Stamens five, free from each other, **epipetalous** ($\overbrace{\mathrm{C}_{(5)}\ \mathrm{A}_5}$).
- **Gynoecium: Bicarpellary, obligately (obliquely) placed, syncarpous**; **ovary superior** ($\underline{\mathrm{G}_{(2)}}$), bilocular, **placenta swollen with many ovules**, **axile placentation**.
- **Fruits:** Berry or capsule.
- **Seeds:** Many, **endospermous**.

Family Solanaceae Floral Formula:

$$\oplus\ \hermaphrodite\ \mathrm{K}_{(5)}\ \overbrace{\mathrm{C}_{(5)}\ \mathrm{A}_5}\ \underline{\mathrm{G}_{(2)}}$$

Economic Importance of Solanaceae (High-Yield NEET Table):

Category	Plants / Examples
Source of Food	Tomato, brinjal, potato (*Solanum tuberosum*)
Spice	Chilli
Medicine	**Belladonna** (*Atropa belladonna*), ***Ashwagandha*** (*Withania somnifera*)
Fumigatory	**Tobacco** (*Nicotiana tabacum*)
Ornamentals	**Petunia**

8. Worked Examples (Conceptual Application)

Worked Example 1: Identifying Simple vs. Compound Leaves

Problem: A botany student examines two twigs. Twig A has a single large blade with deep incisions that do not reach the midrib, and a bud in the axil of its petiole. Twig B has 12 small green blade-like structures arranged laterally along an axis, with a bud present only at the base of the main axis where it joins the stem, and no buds at the base of the 12 small blades. Identify the leaf types in Twig A and Twig B and name the main axis in Twig B.

Solution:

- In **Twig A**, although the lamina is incised, the incisions do not touch the midrib, and an axillary bud is present in the axil of the petiole. Therefore, Twig A bears a **simple leaf**.
- In **Twig B**, the incisions have reached the midrib, breaking the lamina into 12 **leaflets**. A bud is present in the axil of the main petiole, but *not* in the axil of the individual leaflets. Because the leaflets are borne laterally on a common axis, it is a **pinnately compound leaf** (as seen in neem), and the common axis representing the midrib is called the **rachis**.

Worked Example 2: Decoding Floral Symmetry, Aestivation, and Stamen Cohesion

Problem: Classify the following plants—*China rose, Pea, Mustard, and Cassia*—according to (a) Floral symmetry, (b) Corolla aestivation, and (c) Condition of stamens (where applicable from standard chapter facts).

Solution:

- **China rose:** (a) Actinomorphic (or hypogynous superior ovary), (b) **Twisted** aestivation, (c) **Monoadelphous** stamens (united into one bundle).
- **Pea:** (a) **Zygomorphic** ($\%$), (b) **Vexillary (papilionaceous)** aestivation (standard, wings, keel), (c) **Diadelphous** stamens (two bundles).
- **Mustard:** (a) **Actinomorphic** ($\oplus$), (b) Valvate corolla (C_4), (c) Stamens in two whorls (A_{2+4}) showing **variation in the length of filaments** within the flower.
- ***Cassia*:** (a) **Zygomorphic** ($\%$), (b) **Imbricate** aestivation (overlapping in no particular direction).

Worked Example 3: Distinguishing Ovary Chambering in Placentation Types

Problem: Distinguish between the internal chambering (locules) of the ovary in (i) Tomato, (ii) Mustard, and (iii) *Dianthus*.

Solution:

- **(i) Tomato:** Shows **axile placentation**. The ovary is truly **multilocular** (partitioned by true septa), and ovules are attached to the central axial placenta.
- **(ii) Mustard:** Shows **parietal placentation**. Ovules develop on the inner wall/periphery. The ovary is originally one-chambered (unilocular) but becomes **two-chambered (bilocular) due to the formation of a false septum**.
- **(iii) *Dianthus*:** Shows **free central placentation**. Ovules are borne on a central axis and **septa are completely absent** (unilocular ovary).

9. NEET UG Practice Questions with Worked Solutions

Note: All questions below are original exam-aligned practice questions designed strictly for NEET UG preparation based on the chapter syllabus.

Practice Question 1 • Single-Choice MCQ

Which of the following statements is **correct** regarding the regions of a growing root-tip?

(A) The region of meristematic activity is located proximal to the region of elongation.

(B) Root hairs arise from the epidermal cells of the region of elongation to maximize water absorption.

(C) Cells of the region of meristematic activity are very small, thin-walled, and possess dense protoplasm.

(D) The region of maturation is primarily responsible for the growth of the root in length.

Correct Answer: (C)

Worked Solution: A few millimetres above the root cap lies the region of meristematic activity, whose cells are very small, thin-walled, with dense protoplasm, and divide repeatedly. Option (A) is incorrect because the region of elongation is proximal to the meristematic region. Option (B) is incorrect because root hairs arise from epidermal cells of the *region of maturation*. Option (D) is incorrect because the *region of elongation* is responsible for the growth of the root in length.

Practice Question 2 • Single-Choice MCQ

Adventitious roots—roots that arise from parts of the plant other than the radicle—are found in all of the following plants **except**:

(A) Grass

(B) *Monstera*

(C) Banyan tree

(D) Mustard

Correct Answer: (D)

Worked Solution: In dicotyledonous plants such as **mustard**, direct elongation of the radicle leads to the formation of the primary root and its lateral branches, constituting the **tap root system**. Grass, *Monstera*, and the banyan tree are classic examples of plants bearing adventitious roots.

Practice Question 3 • Single-Choice MCQ

Identify the **correct** match between the phyllotaxy type and its corresponding set of plant examples:

(A) Alternate — China rose, *Calotropis*, sunflower

(B) Opposite — Guava, *Calotropis*

(C) Whorled — Silk cotton, *Alstonia*

(D) Opposite — Mustard, guava, china rose

Correct Answer: (B)

Worked Solution: In opposite phyllotaxy, a pair of leaves arise at each node and lie opposite to each other, as seen in ***Calotropis* and guava**. Alternate phyllotaxy is seen in china rose, mustard, and sunflower. Whorled phyllotaxy (more than two leaves at a node) is seen in *Alstonia* (silk cotton is an example of a palmately compound leaf, not whorled phyllotaxy).

Practice Question 4 • Assertion–Reason

Assertion (A): In a cymose inflorescence, the main axis shows limited growth and flowers are borne in a basipetal order.
Reason (R): In cymose inflorescence, the main axis terminates in a flower.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) Both (A) and (R) are false.

Correct Answer: (A)

Worked Solution: In cymose inflorescence, the shoot apical meristem of the main axis is consumed in forming a terminal flower. Because the apex terminates in a flower, the main axis has limited growth, and younger flowers develop laterally below it in a basipetal succession. Thus, (R) directly explains (A).

Practice Question 5 • Single-Choice MCQ

Which of the following sets of plants possesses **perigynous flowers** with a **half-inferior ovary**?

(A) Mustard, china rose, brinjal

(B) Guava, cucumber, ray florets of sunflower

(C) Plum, rose, peach

(D) Pea, bean, *Cassia*

Correct Answer: (C)

Worked Solution: When the gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, the flower is called **perigynous** and the ovary is **half-inferior**. Standard examples are **plum, rose, and peach**. (Mustard, china rose, and brinjal are hypogynous; guava, cucumber, and ray florets of sunflower are epigynous).

Practice Question 6 • Matching Type

Match the type of aestivation in **Column I** with its characteristic description and example in **Column II**:

Column I (Aestivation)	Column II (Description & Example)
P. Valvate	1. Overlapping in no particular direction (*Cassia*, gulmohur)
Q. Twisted	2. Margins just touch one another without overlapping (*Calotropis*)
R. Imbricate	3. Standard overlaps wings, which overlap keel (Pea, bean)
S. Vexillary	4. One margin overlaps that of the next one regularly (China rose, cotton)

(A) P–2, Q–4, R–1, S–3

(B) P–2, Q–1, R–4, S–3

(C) P–4, Q–2, R–1, S–3

(D) P–3, Q–4, R–2, S–1

Correct Answer: (A)

Worked Solution: Valvate (P–2) shows non-overlapping touching margins as in *Calotropis*. Twisted (Q–4) shows regular directional overlapping as in china rose, lady's finger, and cotton. Imbricate (R–1) shows non-directional overlapping as in *Cassia* and gulmohur. Vexillary/papilionaceous (S–3) has 5 petals (standard, two wings, two keel) as in pea and bean.

Practice Question 7 • Multi-Statement Evaluation

Read the following statements regarding the androecium in flowering plants and select the option containing **only correct statements**:

I. A sterile stamen is called a staminode.

II. In lily flowers, stamens are epipetalous.

III. Stamens are monoadelphous in china rose, diadelphous in pea, and polyadelphous in *Citrus*.

IV. There is a variation in the length of filaments within a flower in *Salvia* and mustard.

(A) I, II, and III only

(B) I, III, and IV only

(C) II and IV only

(D) I, II, III, and IV

Correct Answer: (B)

Worked Solution: Statements I, III, and IV are completely accurate. Statement II is incorrect because in lily, the calyx and corolla are not distinct (forming a perianth), so stamens attached to the perianth are termed **epiphyllous** (whereas epipetalous stamens are seen in brinjal).

Practice Question 8 • Single-Choice MCQ

Match the placentation types with their correct examples:

(A) Parietal — *Dianthus* and Primrose

(B) Free central — Mustard and *Argemone*

(C) Axile — China rose, tomato, and lemon

(D) Basal — Pea and bean

Correct Answer: (C)

Worked Solution: Axile placentation (axial placenta in a multilocular ovary) occurs in **china rose, tomato, and lemon**. Mustard and *Argemone* have parietal placentation; *Dianthus* and Primrose have free central placentation; sunflower and marigold have basal placentation; pea has marginal placentation.

Practice Question 9 • Single-Choice MCQ

How does the drupe fruit of **coconut** differ morphologically from the drupe fruit of **mango**?

(A) Coconut develops from a multicarpellary inferior ovary, whereas mango develops from a monocarpellary superior ovary.

(B) In coconut, the mesocarp is fibrous, whereas in mango, the mesocarp is fleshy and edible.

(C) Coconut lacks a stony hard endocarp, whereas mango possesses a stony endocarp.

(D) Mango is a parthenocarpic berry, whereas coconut is a true drupe.

Correct Answer: (B)

Worked Solution: Both mango and coconut are **drupes** that develop from monocarpellary superior ovaries and are one-seeded, with a stony hard endocarp. However, in mango the middle **mesocarp is fleshy and edible**, whereas in coconut the **mesocarp is fibrous**.

Practice Question 10 • Single-Choice MCQ

Which of the following pairs correctly identifies an **endospermic dicotyledonous seed** and a **non-endospermic monocotyledonous seed**, respectively?

(A) Pea and Maize

(B) Castor and Orchid

(C) Gram and Wheat

(D) Bean and Castor

Correct Answer: (B)

Worked Solution: Most dicot seeds (bean, gram, pea) are non-endospermous, but **castor** is an important exception—it is an **endospermic dicot seed**. Conversely, most monocot seeds (maize, wheat) are endospermic, but **orchids** are an important exception—they have **non-endospermic monocot seeds**.

Practice Question 11 • Single-Choice MCQ

In a cereal grain such as maize, the single large shield-shaped cotyledon and the proteinous layer separating the endosperm from the embryo are called, respectively:

(A) Coleoptile and Scutellum

(B) Scutellum and Aleurone layer

(C) Coleorhiza and Tegmen

(D) Pulvinus and Aleurone layer

Correct Answer: (B)

Worked Solution: In monocotyledonous seeds like maize, the embryo has one large, shield-shaped cotyledon known as the **scutellum**, and the outer covering of the bulky endosperm separates the embryo by a proteinous layer called the **aleurone layer**.

Practice Question 12 • Single-Choice MCQ

Examine the floral formula given below:

$$\oplus \ \hermaphrodite \ \mathrm{K}_{(5)} \ \overbrace{\mathrm{C}_{(5)} \ \mathrm{A}_{5}} \ \underline{\mathrm{G}}_{(2)}$$

Which of the following sets of plants belongs to the family represented by this floral formula?

(A) Mustard, cabbage, radish

(B) Pea, bean, gulmohur, *Cassia*

(C) Belladonna, *ashwagandha*, petunia, tobacco, chilli

(D) Lily, wheat, maize, orchid

Correct Answer: (C)

Worked Solution: The floral formula $\oplus \ \hermaphrodite \ \mathrm{K}_{(5)} \ \overbrace{\mathrm{C}_{(5)} \ \mathrm{A}_{5}} \ \underline{\mathrm{G}}_{(2)}$ represents **Family Solanaceae** (actinomorphic, bisexual, 5 united persistent sepals, 5 united petals, 5 epipetalous stamens, bicarpellary syncarpous superior ovary). Economic members of Solanaceae include tomato, brinjal, potato (food), chilli (spice), belladonna and *ashwagandha* (medicine), tobacco (fumigatory), and petunia (ornamental).

10. Common Pitfalls & Rapid Revision Checklist

Common NEET UG Pitfalls to Avoid

- **Bud Position in Compound Leaves:** Never say leaflets bear axillary buds! An axillary bud is present in the axil of the petiole in *both* simple and compound leaves, but **never** in the axil of leaflets.
- **Acropetal vs. Basipetal Order:** Leaves on a stem and flowers in a **racemose** inflorescence are arranged in **acropetal** order. Flowers in a **cymose** inflorescence are arranged in **basipetal** order.
- **Apocarpous vs. Syncarpous Examples:** *Lotus and rose* are **apocarpous** (free carpels), whereas *mustard and tomato* are **syncarpous** (fused carpels).
- **Seed Endosperm Exceptions:** Do not assume all dicots are non-endospermic or all monocots are endospermic. **Castor** is an *endospermic dicot*; **Orchid** is a *non-endospermic monocot*; and seeds of **Family Solanaceae** (a dicot family) are also **endospermous**!
- **Epipetalous vs. Epiphyllous:** Brinjal (Solanaceae) has **epipetalous** stamens ($\overline{\mathrm{C\ A}}$), whereas lily has **epiphyllous** stamens ($\overline{\mathrm{P\ A}}$).
- **Mother Axis in Floral Diagram:** The dot at the top of a floral diagram represents the position of the **mother axis** (posterior side), *not* the bract (which is drawn at the bottom/anterior side if present).

Final Master Summary of Examples (One-Look NEET Revision Table)

Morphological Category	Sub-type	Exact Chapter Examples
Roots	Tap root	Mustard
	Fibrous root	Wheat
	Adventitious root	Grass, *Monstera*, banyan tree
Leaves & Phyllotaxy	Pinnately compound	Neem
	Palmately compound	Silk cotton
	Alternate phyllotaxy	China rose, mustard, sunflower
	Opposite phyllotaxy	*Calotropis*, guava
	Whorled phyllotaxy	*Alstonia*
Symmetry	Actinomorphic ($\oplus$)	Mustard, *Datura*, chilli
	Zygomorphic ($\%$)	Pea, gulmohur, bean, *Cassia*
	Asymmetric	*Canna*
Ovary Position	Hypogynous (Superior, $\underline{\mathrm{G}}$)	Mustard, china rose, brinjal
	Perigynous (Half-inferior)	Plum, rose, peach
	Epigynous (Inferior, $\overline{\mathrm{G}}$)	Guava, cucumber, ray florets of sunflower
Aestivation	Valvate	*Calotropis*
	Twisted	China rose, lady's finger, cotton
	Imbricate	*Cassia*, gulmohur
	Vexillary (Papilionaceous)	Pea, bean
Androecium	Epipetalous	Brinjal
	Epiphyllous	Lily
	Monoadelphous	China rose
	Diadelphous	Pea
	Polyadelphous	*Citrus*
	Unequal filament length	*Salvia*, mustard
Placentation	Marginal	Pea
	Axile	China rose, tomato, lemon
	Parietal (false septum)	Mustard, *Argemone*
	Free central (no septa)	*Dianthus*, Primrose
	Basal (single ovule)	Sunflower, marigold